

FIELD-SCALE SPATIAL AND TEMPORAL SOIL WATER VARIABILITY IN IRRIGATED CROPLANDS



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HIGHLIGHTS

- The greatest heterogeneity in soil water was observed at 30-60 cm depth in a corn field and 0-15 cm in a cotton field.
- Spatiotemporal soil water variability did not increase with increasing soil water in all soil layers in both croplands during the growing season.
- Water excess and water stress locations were identified in both fields. A single, temporally stable location was identified in each field, which can be used for precise uniform irrigation.
- Knowledge of variability and stability in soil water can be useful in determining the number and location of sensors to install in crop fields to assist irrigation scheduling decisions.

ABSTRACT. *This study investigated the spatiotemporal variability and temporal stability of soil water at various depths in two croplands sown in corn and cotton during the 2018 growing season in the Tennessee Valley Region (TVR) of northern Alabama. Classical statistics and relative difference approaches were used to analyze soil water data in this study. In the corn field, the 30-60 cm depth showed the greatest variability, while the cotton field showed the greatest variability at 0-15 cm depth. A decreasing trend was noticed between mean soil water and coefficient of variation for all depths in the cotton field and at 30-60 cm depth in the corn field. However, convex upward, exponential decreasing, or no trends were found between mean soil water and standard deviation at different depths in both fields. The temporal stability analysis showed one representative sensor (S8 in corn and S1 in cotton) for the entire soil profile in both fields. Different statistical tests, i.e., Spearman's correlation (r_s), Nash-Sutcliffe efficiency (NSE), coefficient of determination (R^2), etc., were used to reduce uncertainty or increase confidence in the performance of representative sensors. Among various field attributes, topography in corn and soil properties in cotton were determined as significant factors responsible for soil water variability. Crop evapotranspiration (ET_c) showed significant negative weak and moderate correlations with soil water in the corn and cotton fields, respectively. However, the mean air temperature showed a significant positive correlation with soil water in the corn field and a significant negative correlation in the cotton field. Solar radiation had a significant negative correlation with soil water in the cotton field and a non-significant correlation in the corn field. Accumulated growing degree days (accGDD) showed a significant negative correlation with soil water in the corn field and a positive correlation in the cotton field. This study gives insights into soil water variability, provides useful information about temporal stability, and identifies significant factors for precision uniform irrigation scheduling.*

Keywords. *Corn, Cotton, Croplands, Growing season, Irrigation, Soil water, Spatial and temporal variability, Temporal stability.*

Understanding spatial and temporal soil water dynamics in croplands is of utmost importance for irrigation scheduling and for understanding surface and subsurface transport of water, nutrients, and contaminants (Filippucci et al., 2020; Gao et al., 2019; Su et al., 2018). The spatial and temporal variations in soil

water have been studied for decades around the globe at plot to watershed scales in different natural landscapes (Brocca et al., 2009; Hao et al., 2020; Shen et al., 2016). In many hydrological studies, the temporal stability concept has been used extensively to calibrate hydrological models, for flood forecasting, and for validating remote sensing data in natural landscapes (Brocca et al., 2018; Hao et al., 2020; Yetbarek and Ojha, 2020; Zaussinger et al., 2019). Knowledge of soil water variability can be helpful in erosion control and irrigation water management in crop fields during a growing season (Yetbarek and Ojha, 2020). However, plot-scale studies

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always have a limitation of uncertainty with upscaling to field scale (Kornelsen and Coulibaly, 2013), and studies at large scale (watershed) and small scale (plot) are of limited use to farmers. Hupet and Vanclooster (2002) suggested a need for more research at the field scale in different crop fields.

Researchers have investigated the relationship between average soil water and standard deviations to understand spatial variability across time at various depths. However, relatively few studies are available for deeper soil profiles, as compared to those that have studied surface soil water, especially in croplands (Bauer-Marschallinger et al., 2019; Gao and Shao, 2012; She et al., 2012, 2015). There is no perfect way to explain soil water variability, and studies have shown positive, negative, and neutral correlations between average soil water and standard deviations in natural agricultural vegetation (Guber et al., 2008; Shen et al., 2016; Vereecken et al., 2014). Due to the lack of accuracy and precision in satellite data at the field scale, *in situ* soil water measurements and irrigation observations are necessary to understand soil water variability within the soil profile of croplands during the growing season (Filippucci et al., 2020). Rainfed agricultural practices are common in southeastern U.S. states such as Alabama and Tennessee. However, irrigation adoption is increasing, especially in Alabama, where irrigation has increased by 25% from 2012 to 2017 (NASS, 2017). Farmers are adopting irrigation to increase crop yields, avoid water stress, and increase water use efficiency during the growing season (FAO, 2018). Climatic variations directly influence the spatial and temporal distribution of precipitation, which challenges farmers to schedule precise and accurate irrigation. A better understanding of soil water variability in crop fields is needed to schedule precise irrigation.

Although irrigation adoption is increasing in the U.S., the main challenges to irrigation adoption are financial (NASS, 2013). Therefore, farmers schedule irrigation based on the feel of the soil and by looking at the crop at different times during the growing season (NASS, 2013, and according to a farmer survey in the TVR). Irrigation is an essential part of the hydrological cycle and food production. It accounts for 70% or more of freshwater withdrawals globally (Filippucci et al., 2020). Water withdrawals for irrigation are a concern for water availability in areas where water resources are limited. Improvements in irrigation efficiency would not reduce costs enough to cover the sensor installation costs and reduce uncertainty about the future availability of water (NASS, 2013), which motivated us to identify representative locations of soil water variability to install sensors in croplands.

Accurate estimation of spatial mean soil water is essential in crop fields because irrigation scheduling is often based on soil water measured at a single point or just a few points (Hao et al., 2020). Therefore, it is necessary to study soil water variability in crop fields to help farmers identify water stress, water excess, and temporally stable locations to estimate irrigation amounts and schedule precision irrigation during the growing season (Hao et al., 2020; Yetbarek and Ojha, 2020). Soil water products, such as ASCAT (Advanced Scatterometer) and SMAP (Soil Moisture Active Passive), have been used to produce soil water data, but they

cannot be used in crop fields due to low resolution (Palombo et al., 2019). Therefore, it is important to study soil water variability in crop fields to help farmers improve their water use efficiency and understand the significant factors for irrigation scheduling during the growing season.

Additionally, it is important to collect *in situ* soil water measurements in deeper layers to assist in hydrological modeling and validation of climate models, especially in croplands (Brocca et al., 2010; Heathman et al., 2009; Loew et al., 2013). Although *in situ* soil water measurements are expensive, time-consuming, and laborious, there is a great need in scientific and farming communities to understand the depth-dependent spatial and temporal variability of soil water in croplands, especially during growing seasons (Majdar et al., 2018). Certain locations might be temporally stable for soil water and can be used as representative locations (Vachaud et al., 1985) for irrigation scheduling decisions. Many studies have been carried out to understand temporal stability in soil water using *in situ* measurements, remote sensing, or a combination of both methods at various depths and scales to find representative locations for hydrologic modeling purposes (Dari et al., 2019; She et al., 2015; Shen et al., 2016). According to temporal stability investigations, if a location is wet at one time, it will remain wet another time if the spatial pattern of temporal variation does not vary with time (Vachaud et al., 1985). However, a significant challenge for irrigation scheduling is the spatial variation in soil water and the difficulty in obtaining representative locations for prescribing irrigation. Efficient site-specific management of soil water with adequate irrigation requires temporal stability analysis of soil water (Yetbarek and Ojha, 2020). Irrigation scheduling using a single, temporally stable sensor that represents the field soil water can be useful and cost-effective to prescribe uniform irrigation.

Various crucial factors, including terrain attributes, soil properties, vegetation cover, and meteorological parameters, affect soil water variability (Beven and Kirkby, 1979; Mei et al., 2018; Takagi and Lin, 2012). Topographic factors have been studied more extensively than vegetation and meteorological factors. Limited research has been done at the field scale under different types of vegetation, especially in croplands (Fry and Guber, 2020; Guber et al., 2008; Rossato et al., 2017; Yetbarek and Ojha, 2020), due to the extensive work required in crop fields during a growing season. Different soil textures have shown varying degrees of temporal stability and variability of soil water. According to Mohanty and Skaggs (2001), silty loam soil has less stability than sandy loam, and Choi and Jacobs (2011) found that temporal stability decreased with increasing sand content. However, it is still not clear what factors dominate soil water variability in crop fields, especially during a growing season.

The findings of this study can be useful in field-scale hydrological modeling (e.g., Soil Water Assessment Tool (SWAT) and Agricultural Policy/Environmental eXtender (APEX)), for validating remote sensing data, and for calibrating subsurface hydrology models (HYDRUS) in different types of vegetation (Filippucci et al., 2020; Parinussa et al., 2014; Zaussinger et al., 2019). However, many questions still need to be answered, such as: How are spatial and temporal variability affected by uniform irrigation in different

croplands? What are the main factors involved in the spatial variability and temporal stability of soil water in croplands during the growing season under uniform irrigation? How can farmers adapt water use efficient, uniform irrigation in crop fields during the growing season?

Different approaches, such as geostatistical analysis, have been used in many studies to understand soil water variability (Gia Pham et al., 2019; Pandey and Pandey, 2010; Vasu et al., 2017). However, a simple approach of exploratory data analysis can be useful to establish the relationship between spatial mean soil water and its variability. In this study, two crop fields (corn and cotton) were studied for soil water variability using *in situ* soil water measurements to a depth of 60 cm under uniform irrigation management. The objectives of this study were to (1) determine the spatiotemporal variability of soil water in both croplands at different depths, (2) identify representative locations of soil water measurement for scheduling irrigation in the croplands during the growing season, and (3) identify the factors affecting the spatiotemporal variability of soil water in the croplands during the growing season.

MATERIALS AND METHODS

STUDY SITE DESCRIPTION

The study areas, which belonged to farmers, were located in the Tennessee Valley Region (TVR) of northern Alabama (fig. 1). The corn field was approximately 120 ha (a 60 ha area in the middle of the field was used in this study with a slope variation of 0% to 10%), and the cotton field was 12 ha

with a slope variation of 0% to 3%. The corn field (34° 43' 7.15" N, 87° 23' 9.36" W), with minimum and maximum elevations of 168 and 178 m, respectively, above mean sea level, was located in Lawrence County, Alabama, and the cotton field (34° 43' 18.10" N, 87° 1' 16.14" W), with minimum and maximum elevations of 181 and 185 m, respectively, above mean sea level, was located in Limestone County, Alabama. The average annual precipitation in the region (1895 to 2019) is about 1400 mm (Sangha et al., 2020a; https://sercc.com/climateinfo/monthly_seasonal). The average annual precipitation during the growing season (April to October) is 759 mm. The average annual temperature (1895 to 2019) is 17.3°C, with a maximum temperature of 29.1°C in August and a minimum temperature of 13.8°C in December. The major crops grown in the TVR are corn (*Zea mays* L.), cotton (*Gossypium hirsutum* L.), soybean (*Glycine max* (L.) Merr.), and winter wheat (*Triticum aestivum* L.) (Sangha et al., 2020b).

The corn field was seeded on 10 April 2018 and harvested on 3 September 2018. The cotton field was seeded on 2 May 2018 and harvested on 24 October 2018. Both fields were under no-till management. The cotton field was planted with a cover crop before the cotton; however, no cover crop was used in the corn field. According to SSURGO (2020), the dominant soil types in the corn field were Abernathy-Emory silt loams (38%), Decatur silt loam (29%), Decatur silty clay loam (20%), and Decatur silty clay (8%). In the cotton field, the dominant soil types were Guthrie silt loam (25%), Decatur silt loam (24%), Lawrence silt loam (21%), Sango silt loam (14%), and Dickson silt loam (11%). Soil samples were

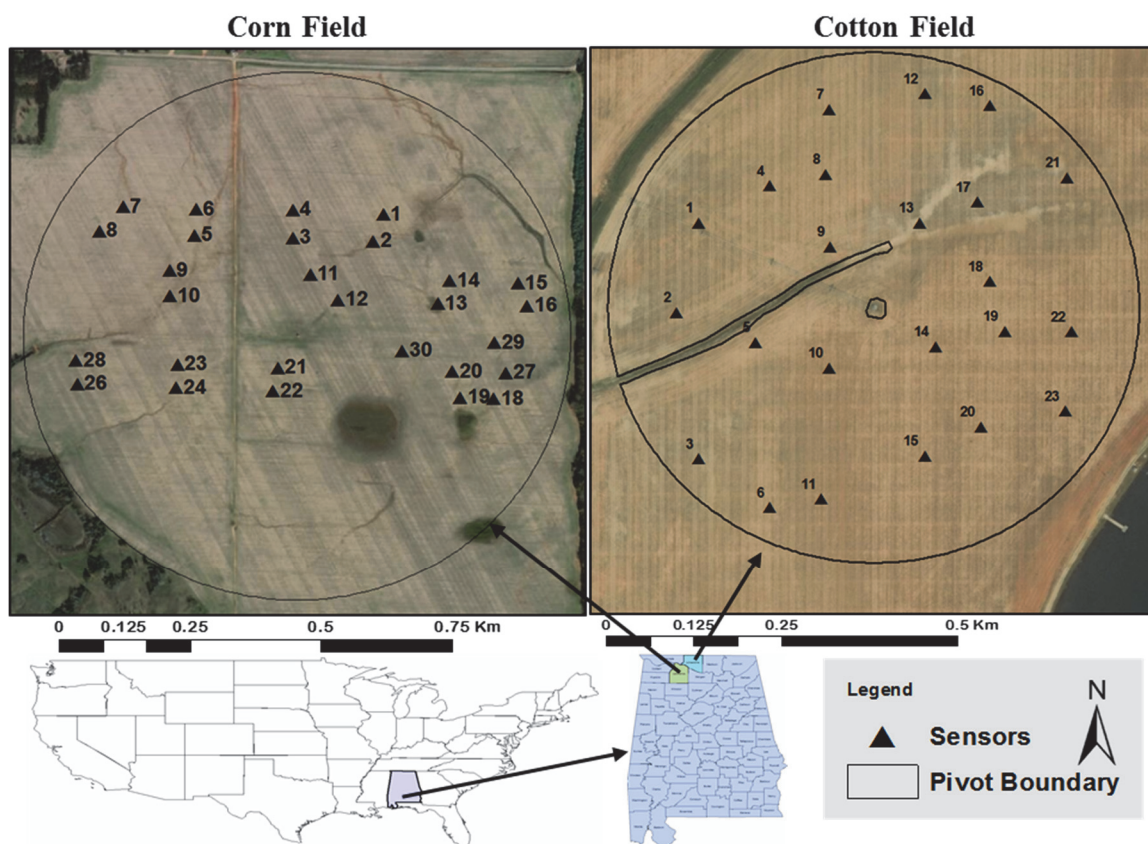


Figure 1. Study sites (corn and cotton fields) showing sensor locations and center-pivot irrigation boundaries during the growing season.

collected in both fields to determine soil texture using the hydrometer method (Bouyoucos, 1962). In the corn field, the soil texture was classified as clay loam for the upper two soil layers (0-15 cm and 15-30 cm) and clay for the deepest (30-60 cm) soil layer. Silty clay loam was the soil texture in the entire soil profile of the cotton field.

FIELD MEASUREMENTS

Both fields were instrumented with Watermark sensors (Irrometer Co., Riverside, Cal.), combined into a wireless soil sensor probe, to measure soil matric potential vertically at 0-15 cm, 15-30 cm, and 30-60 cm depths. Watermark sensors are electrical resistance devices that measure soil water matric potential. Watermark sensors with a similar array have been used in corn and cotton crops for irrigation scheduling at 60 cm depth in the southeastern U.S. (Lena et al., 2020; Liang et al., 2016; Vellidis et al., 2008). Therefore, we adopted a similar array in this study. In addition, greater than 50% of total roots have been found within the top 60 cm depth in corn fields (Anderson, 1987; Fan et al., 2016), and maximum root length density in cotton is found in the top 60 cm of the soil profile (Aggarwal et al., 2017). The sensors were controlled remotely to collect data and to monitor their performance.

Sensor locations were selected based on the farmers' experience, visual conditions in the fields, soil texture, topography, and ten years of historical yield data to capture soil water variability in crop fields and to find temporal stability in soil water measurements to determine an irrigation strategy during the growing season. Initially, 28 and 23 sensors, respectively, were installed in the corn and cotton fields. However, due to errors in measurement, bursting of sensors, and other problems, only 17 and 18 sensors, respectively, in the corn and cotton fields were considered in good condition for this study. Hourly soil water matric potential was measured from 16 May to 12 August 2018 in the corn field and from 4 June to 18 October 2018 in the cotton field. These hourly measurements were converted to daily average data at each location for the respective depths (Li et al., 2019).

The soil matric potential data were converted into soil water data using the van Genuchten model (van Genuchten, 1980). To use the van Genuchten model (eq. 1), soil water retention curves were developed for each depth (0-15 cm, 15-30 cm, and 30-60 cm) for both fields:

$$\theta(\psi) = \theta_r + \frac{\theta_s - \theta_r}{\left[1 + (\alpha|\psi|)^n\right]^{1-1/n}} \quad (1)$$

where $\theta(\psi)$ is soil water ($\text{cm}^3 \text{ cm}^{-3}$), ψ is matric potential (cm), θ_r is residual water content ($\text{cm}^3 \text{ cm}^{-3}$), θ_s is saturated water content ($\text{cm}^3 \text{ cm}^{-3}$), and α and n are shape parameters estimated from the soil water retention curve.

Undisturbed soil samples were collected at each depth to develop soil water retention curves using a hydraulic property analyzer (HYPROP 2, Meter Group, Pullman, Wash.). The intact soil cores were saturated for at least 24 h before running the experiment. Disturbed soil samples were collected to measure permanent wilting point using a dew point

hygrometer (WP4C PotentiaMeter, Meter Group) (Lena et al., 2020).

The fields were equipped with remotely controlled center-pivot irrigation systems. The corn field was equipped with a 625 m Reinke center-pivot irrigation system (Reinke Manufacturing Co., Inc., Deshler, Neb.), and the cotton field was equipped with a 195 m Valley center-pivot irrigation system (Valmont Irrigation, Valley, Neb.). Both irrigation systems had a mid-elevation spray application (MESA) sprinklers equipped with pressure regulators. The center-pivots were operated by the farmers to schedule irrigation during the growing season. Although the sensors and center-pivots were controlled remotely, the research team often visited the fields to verify the irrigation operations and to repair the sensors.

WEATHER DATA

Due to spatial and temporal variation in precipitation (Kumar, 2016; Ridolfi et al., 2020), an automatic weather station (Vantage Pro2 Plus, Davis Instruments, Hayward, Cal.) was installed near each field to record meteorological parameters. Precipitation, temperature, and solar radiation were recorded at 15 min intervals and converted to a daily time step. Evapotranspiration (ET_o) was measured at hourly intervals and converted to a daily time step. The annual precipitation recorded in the corn field in 2018 was 1622 mm, with a maximum monthly precipitation of 285 mm in February and a minimum of 22 mm in July. The total precipitation recorded in the corn field was 582 mm during the 2018 growing season (April to October). The annual precipitation recorded in the cotton field in 2018 was 1679 mm, with a maximum monthly precipitation of 249 mm in February and a minimum of 48 mm in January. The total precipitation recorded in the cotton field was 617 mm during the 2018 growing season. According to historical records (2006 to 2018) of average precipitation, the 2018 growing season was considered a wet season.

Crop evapotranspiration (ET_c) was determined using the FAO-56 method for both fields (eq. 2) (Allen et al. 1998):

$$ET_c = K_c \times ET_o \quad (2)$$

The growing degree days (GDD)-determined growth stages and associated crop coefficients (K_c) estimated using a smart irrigation application for cotton were used to calculate ET_c (Vellidis et al., 2016). The growth stages were assessed for corn during field visits to determine the associated K_c used to calculate ET_c (Irmak, 2017). The GDD was estimated using temperature measurements (Anandhi, 2016). Figure 2 shows precipitation and irrigation with ET_c data in both fields. Figure 3 shows cumulative water input (sum of precipitation and irrigation) and loss (ET_c) in both fields during the growing season. The total amount of irrigation applied was 66.5 mm (five irrigations) in corn and 30.5 mm (two irrigations) in cotton during the growing season.

METHODS OF SOIL WATER VARIABILITY ANALYSIS

Various statistical approaches were adopted (Dari et al., 2019; Shen et al., 2016; Vachaud et al., 1985) to understand

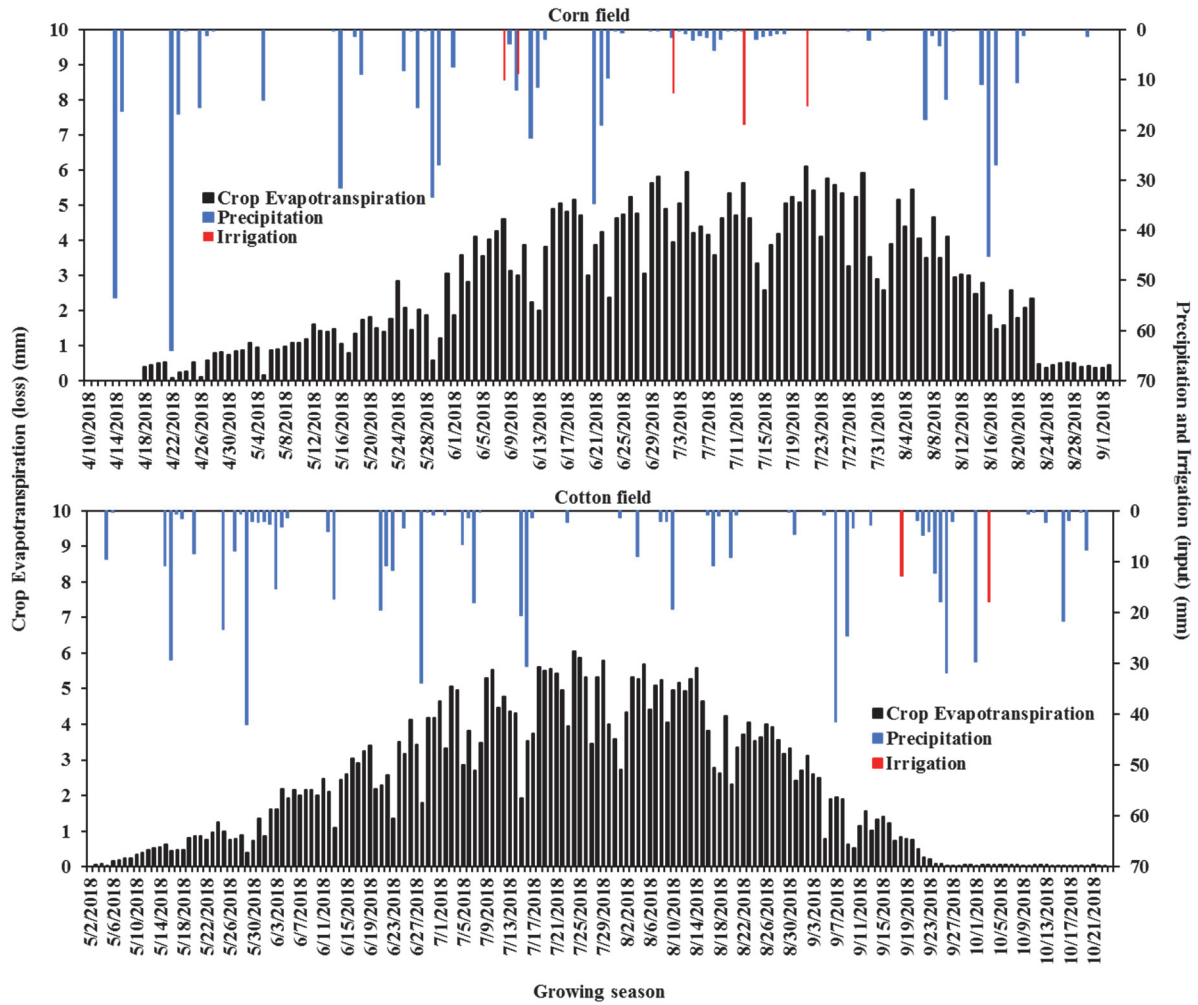


Figure 2. Water input (precipitation and irrigation) and loss (crop evapotranspiration) in corn and cotton fields during the growing season.

the spatial and temporal variability of soil water in both crop fields.

Statistical Analysis

The soil water (θ) measured at each location i ($i = 1, 2, \dots, M$) and on each day j ($j = 1, 2, \dots, N$) was used to determine different statistics for this study, where $M = 17$ and $N = 26$ for the corn field, and $M = 18$ and $N = 56$ for the cotton field. Different statistics were adopted to minimize the uncertainty and increase confidence in the results. The spatial mean ($\bar{\theta}_j$) for each day is given by:

$$\bar{\theta}_j = \frac{\sum_{i=1}^M \theta_{ij}}{M} \quad (3)$$

The temporal mean ($\bar{\theta}_i$) for each location is given by:

$$\bar{\theta}_i = \frac{\sum_{j=1}^N \theta_{ij}}{N} \quad (4)$$

The spatial standard deviation ($\sigma(\theta_j)$) and coefficient of variation ($CV(\theta_j)$) of soil water for each sampling day are given by:

$$\sigma(\theta_j) = \sqrt{\frac{1}{M-1} \sum_{i=1}^M (\theta_{ij} - \bar{\theta}_j)^2} \quad (5)$$

$$CV(\theta_j) = \frac{\sigma(\theta_j)}{\bar{\theta}_j} \quad (6)$$

In the same way, the temporal standard deviation ($\sigma(\theta_i)$) and coefficient of variation ($CV(\theta_i)$) can be found for each location.

Temporal Stability Statistics

Vachaud et al. (1985) introduced a relative difference statistic ($\delta(\theta_{ij})$) to understand the temporal stability of measurement locations. This approach identifies representative locations for a study area based on observed soil water values. All computations are done separately for individual depths.

The relative difference $\delta(\theta_{ij})$ for each location i and each day j can be defined as:

$$\delta(\theta_{ij}) = \frac{(\theta_{ij} - \bar{\theta}_j)}{\bar{\theta}_j} \quad (7)$$

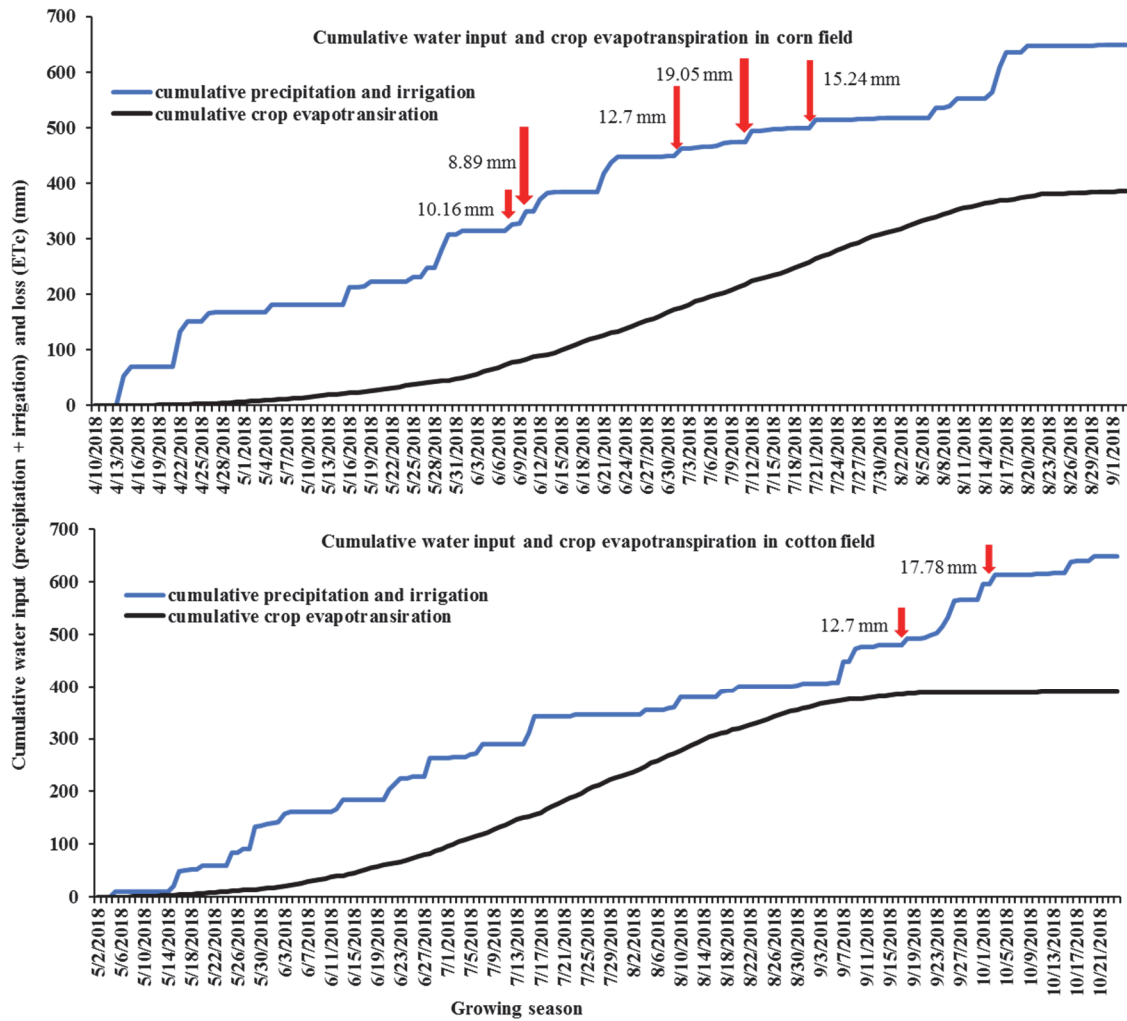


Figure 3. Cumulative water input (precipitation and irrigation) and cumulative crop evapotranspiration in corn and cotton fields during the growing season. Irrigations are indicated by the red arrows.

Based on the relative difference, the mean relative difference ($\bar{\delta}_i$) and standard deviation of relative difference ($\sigma(\delta_i)$) for each location i can be given by:

$$\bar{\delta}_i = \frac{\sum_{j=1}^N \delta(\theta_{ij})}{N} \quad (8)$$

$$\sigma(\delta_i) = \sqrt{\frac{1}{N-1} \sum_{j=1}^N (\delta_{ij} - \bar{\delta}_i)^2} \quad (9)$$

The mean relative difference ($\bar{\delta}_i$) quantifies the departure of soil water at a specific location from the field mean soil water and distinguishes whether the location is drier or wetter than the field mean during a growing period. A representative location shows a mean relative difference of zero and a small standard deviation ($\sigma(\delta_i)$). Values of ($\bar{\delta}_i$) greater or less than zero indicate over- or under-estimation of field mean soil water and indicate wetter and drier conditions, respectively (Jacobs et al., 2004). A location with a small value of $|\bar{\delta}_i|$ and $\sigma(\delta_i)$ is considered representative, or

temporally stable. The mean relative difference is also defined as an index of the temporal variability of soil water.

A single statistic introduced by Jacobs et al. (2004), called the index of time stability (ITS), was used to identify the best sampling location during a growing season. A low ITS value indicates a representative, or temporally stable, location for soil water in a field during the growing season. The ITS for each location is given by:

$$ITS_i = \sqrt{(\bar{\delta}_i^2 + \sigma(\delta_i)^2)} \quad (10)$$

Spearman's rank correlation (r_s) has also been widely used to explain the temporal stability of soil water in croplands. Spearman's rank correlation was computed between the soil water at each location i and the spatial mean soil water on day j during the growing season:

$$r_{s,i} = 1 - \frac{6 \sum_{j=1}^N (R(\theta_{ij}) - R(\bar{\theta}_j))^2}{n(n^2 - 1)} \quad (11)$$

where $R(\theta_{ij})$ is the rank of soil water at location i varying in time j , and $R(\bar{\theta}_j)$ is the rank of mean spatial soil water during time j .

The index of agreement (IA) is a standardized measurement index that was developed by Willmott et al. (1985), also known as the Willmott index, to measure the degree of performance of any statistical analysis. It varies from 0 to 1, where 0 indicates no agreement, and 1 indicates perfect agreement with the data:

$$IA_i = 1 - \frac{\sum_{j=1}^N (\bar{\theta}_j - \theta_{ij})^2}{\sum_{j=1}^N (|\theta_{ij} - \text{mean}(\bar{\theta}_j)| + |\bar{\theta}_j - \text{mean}(\bar{\theta}_j)|)^2} \quad (12)$$

Additional criteria widely used in hydrologic simulations were also adopted to test the temporal stability of soil water and to test if the spatial mean soil water can be represented by one sensor. The goal was to see if these adopted statistics could also provide similar conclusions using R^2 , RMSE, Kling-Gupta efficiency (KGE) (Gupta et al., 2009; Knoben et al., 2019), and Nash-Sutcliffe efficiency (NSE) (Nash and Sutcliffe, 1970):

$$NSE = 1 - \frac{\sum_{j=1}^N (\bar{\theta}_j - \theta_{ij})^2}{\sum_{j=1}^N (\bar{\theta}_j - \text{mean}(\bar{\theta}_j))^2} \quad (13)$$

$$KGE = 1 - \sqrt{(r-1) + (\alpha-1) + (\beta-1)} \quad (14)$$

$$\alpha = \frac{\sigma_{\theta_i}}{\sigma_{\bar{\theta}_j}}, \quad \beta = \frac{\mu_{\theta_i}}{\mu_{\bar{\theta}_j}}$$

where r is the correlation coefficient between the temporally stable sensor and field mean soil water, α is the variability ratio between the standard deviation of each location (σ_{θ_i}) and the standard deviation of spatial mean soil water ($\sigma_{\bar{\theta}_j}$), and β is the ratio between average soil water at each location (μ_{θ_i}) and the average of spatial mean soil water ($\mu_{\bar{\theta}_j}$). The values of KGE and NSE range from $-\infty$ to 1. A value close to 1 indicates a perfect association between the data.

R^2 and RMSE are common and widely used statistics in hydrology; therefore, the equations are not shown here. The statistics were evaluated between the soil water at each location i and the spatial mean soil water on day j in both crop fields during the growing season. This provided another method to verify the temporal stability of the sensor used to represent the field mean soil water in each field. The closer r , R^2 , KGE, and NSE are to 1 and the closer RMSE is to zero, the more accurate the representative sensor is considered for estimating field mean soil water.

The number of sensors required to estimate average soil water within a certain absolute error limit can be determined for the maximum soil water variability through implicit relationships developed by Wang et al. (2008):

$$NRS = t_{1-\alpha/2, NRS-1}^2 \times \left(\frac{(\sigma(\theta_j))^2}{AE^2} \right) \quad (15)$$

where

NRS = degree of freedom

$t_{1-\alpha/2, NRS-1}^2$ = value of t -distribution at α significance

AE = absolute error.

FACTORS RESPONSIBLE FOR SOIL WATER VARIABILITY

Meteorological parameters were used in this study to find correlations with field mean soil water during the growing season. These meteorological parameters were recorded for both fields. Topographic and soil parameters, including elevation, slope, electrical conductivity (EC, measured with a V3100 meter, Veris Technologies, Salina Kans.) at shallow (0-30 cm) and deep (0-90 cm) depths in the soil profile, and sand, silt, and clay percentages were also used to find correlations (if any) with mean soil water. The elevation data were collected using a John Deere StarFire 6000 system (a real-time kinematic GPS with horizontal and vertical accuracy of 5 and 2.5 cm, respectively) installed on the harvesters. The slopes of both fields were estimated using ArcGIS (ver. 10.3.1). Spearman's rank correlation was used to find significant correlations between the parameters and mean soil water in both fields (Majdar et al., 2018) to evaluate the significant factors that need to be considered in irrigation scheduling.

RESULTS AND DISCUSSION

SPATIAL VARIABILITY IN SOIL WATER

Figure 4 shows the soil water in both fields at different depths at various locations during the growing season. The box and whisker plots show variations in soil water at each sensor location with maximum, minimum, median, and upper and lower quartiles. Soil water varied from location to location within each field, and sensor 30 (S30) indicated the minimum average soil water in the entire soil profile in the corn field (figs. 4a to 4c). This variation in soil water from one location to another could be due to differences in soil properties, such as hydraulic conductivity (Heathman et al., 2012). However, due to precipitation and irrigation (Heathman et al., 2012; Shen et al., 2016), the deepest soil layer (30-60 cm) held greater mean soil water, followed by the 15-30 cm and 0-15 cm soil depths in the corn field (figs. 4a to 4c and 5a). In the cotton field, sensor 3 (S3) indicated the minimum average soil water at the 0-15 cm and 30-60 cm depths (figs. 4d to 4f). However, sensor 18 (S18) indicated the maximum average soil water at the 0-15 cm and 15-30 cm depths (figs. 4d to 4f). The variation in soil water with location might be due to differences in soil properties and cover crops in the cotton field (Yetbarek and Ojha, 2020). The least variation in soil water (figs. 4b and 4e) was observed at the 15-30 cm depth in both fields.

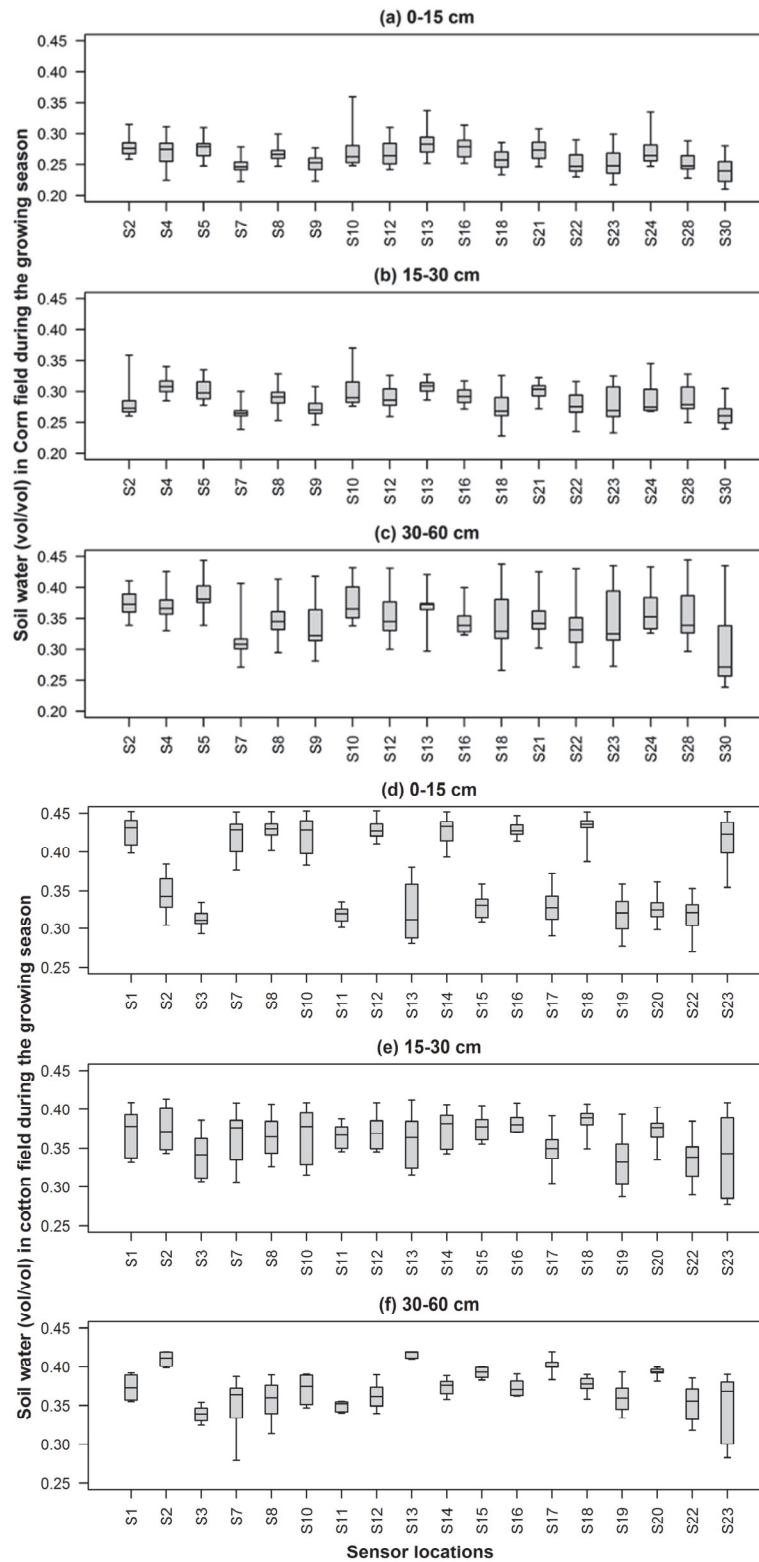


Figure 4. Box plots of soil water variability in the (a-c) corn field and (d-f) cotton field during the 2018 growing season.

The variation in soil water between the two fields could be due to differences in soil hydraulic properties (Heathman et al., 2012), topography, climate, root distribution, and water uptake within the soil profiles (Liu et al., 2018). For example, topography significantly affected soil water in the

corn field, but not in the cotton field (table 1). On the other hand, the silt percentage significantly affected soil water in the cotton field, but not in the corn field.

The variation in spatial mean soil water (fig. 5) at different depths could be linked to the changes in daily

Table 1. Spearman's rank correlation of temporal mean soil water with topographic attributes and soil parameters.^[a]

Factor	Corn Field			Cotton Field		
	0-15 cm	15-30 cm	30-60 cm	0-15 cm	15-30 cm	30-60 cm
Elevation (m)	-0.39	-0.31	-0.45*	0.18	-0.15	-0.23
Slope (%)	-0.20	-0.25	-0.32	0.32	-0.04	-0.10
Sand (%)	0.23	-0.13	-0.25	-0.27	-0.35	-0.08
Silt (%)	-0.06	0.03	0.32	0.50*	0.27	-0.18
Clay (%)	-0.30	0.10	-0.25	-0.01	-0.06	0.16
EC	NA	NA	NA	NA	-0.54*	-0.47*

^[a] Asterisks (*) indicate significance at 0.1 (NA = not available).

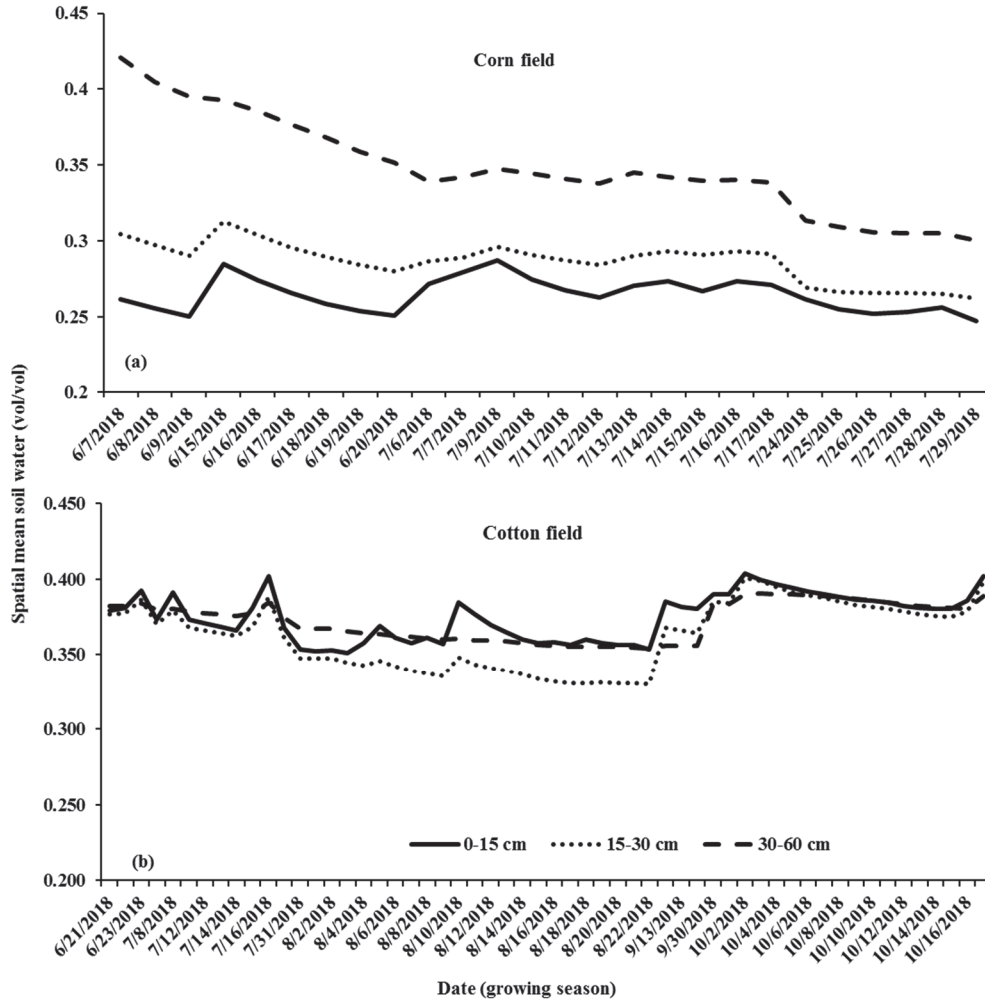


Figure 5. Spatial mean soil water distribution in the (a) corn field and (b) cotton field during the 2018 growing season.

precipitation, irrigation, and ET_c in both fields during the growing season. The fluctuations in standard deviation (fig. 6) at different depths were due to variation in soil water caused by water input and losses. For example, in the corn field on June 15, an increase in soil water (fig. 5a) at the 0-15 cm and 15-30 cm depths can be observed due to rainfall events (fig. 2a, 12 to 14 June), but a decrease occurred at the 30-60 cm depth. The change in standard deviation (fig. 6a) can be seen at the 30-60 cm depth, but not in the other two soil depths. Likewise, the effects of rainfall or irrigation on soil water and standard deviation were noticed in both fields during the growing season. It is a little complicated to understand the trends in soil water and standard deviation separately (figs. 5 and 6). Therefore, we developed relationships

between soil water and standard deviation in soil water at separate depths (fig. 7).

In the corn field, a convex upward trend with a threshold of 0.345 (vol/vol) was detected between $\sigma(\theta_j)$ and $\bar{\theta}_j$ at the 30-60 cm depth. An exponentially decreasing trend between $CV(\theta_j)$ and $\bar{\theta}_j$ at the same depth (fig. 7c for $CV(\theta_j)$ vs. $\bar{\theta}_j$) showed consistency with previous studies at 0-15 cm depth with different land uses (Brocca et al., 2012; Dari et al., 2019). A convex trend explains that the variability in soil water increases and decreases, respectively, before and after a threshold value of soil water. An exponential relationship explains that the variability parameters decrease with increasing soil water. However, no trends in $CV(\theta_j)$ and $\sigma(\theta_j)$

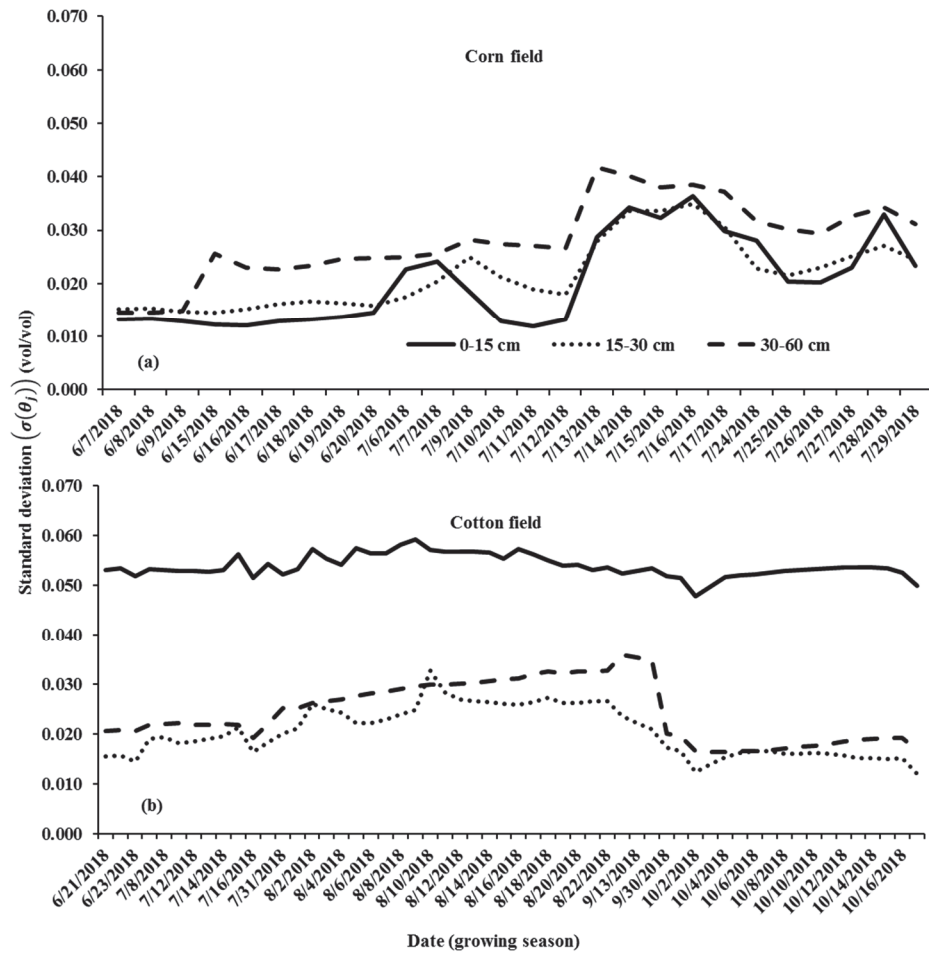


Figure 6. Distribution of standard deviation in the (a) corn field and (b) cotton field at different depths during the 2018 growing season.

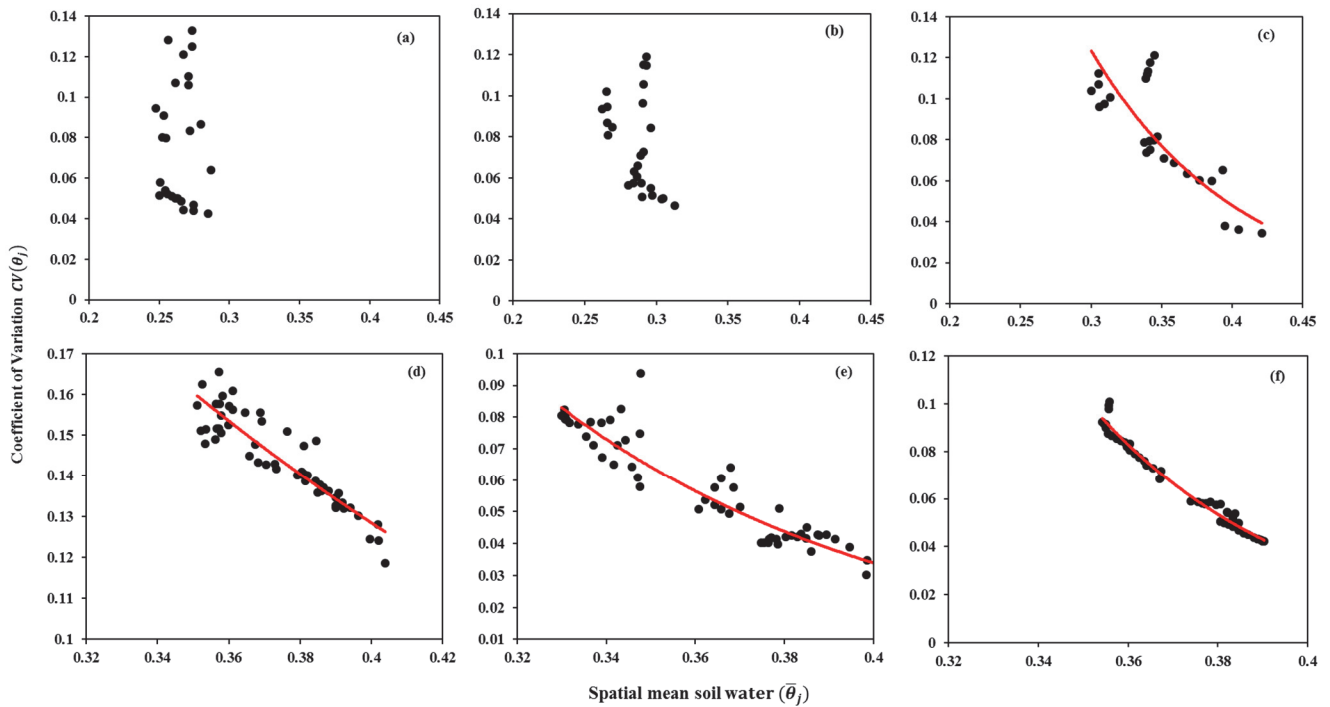


Figure 7. Trend (red lines) between coefficient of variation and spatial mean soil water in the corn field at (a) 0-15 cm, (b) 15-30 cm, and (c) 30-60 cm depths and in the cotton field at (d) 0-15 cm, (e) 15-30 cm, and (f) 30-60 cm depths during the 2018 growing season.

with soil water ($\bar{\theta}_j$) were detected at the 0-15 cm and 15-30 cm depths in the corn field (figs. 7a and 7b for $CV(\theta_j)$ and $\bar{\theta}_j$). Similar trends were observed by Brocca et al. (2010). These patterns have not been reported in previous studies for any croplands.

In the cotton field, a convex upward pattern with a threshold of 0.357 (vol/vol) was detected between $\sigma(\theta_j)$ and $\bar{\theta}_j$ at the 0-15 cm depth, while an exponentially decreasing trend was detected between $CV(\theta_j)$ and $\bar{\theta}_j$ (fig. 7). For the 15-30 cm and 30-60 cm depths, an exponentially decreasing trend of $CV(\theta_j)$ and $\sigma(\theta_j)$ with soil water ($\bar{\theta}_j$) was detected. Similar results regarding the trends in $CV(\theta_j)$ and $\sigma(\theta_j)$ with $\bar{\theta}_j$ were obtained in previous studies with various land uses in the watershed (Brocca et al., 2010; Dari et al., 2019) but not in croplands. The trends are shown using $CV(\theta_j)$ and $\bar{\theta}_j$ plots. Therefore, $\sigma(\theta_j)$ and $\bar{\theta}_j$ plots are not shown.

However, some of the trends in this study did not follow previous studies. For example, a convex trend between $CV(\theta_j)$ and $\bar{\theta}_j$ in the corn field and an exponential trend between $\sigma(\theta_j)$ and $\bar{\theta}_j$ in the cotton field were detected during the growing season. The relationships detected between $\sigma(\theta_j)$ and $\bar{\theta}_j$ in this study did not align with Shen et al. (2016), who showed an increasing trend at different depths in cropland in an arid watershed-scale study with a shallow groundwater table during the growing season. This contradiction in the relationship between watershed and field-scale studies has been noted by others (Brocca et al., 2007, 2010). We found that the cotton field showed a higher standard deviation in soil water than the corn field at the 0-15 cm depth during the growing season. We also found that the maximum standard deviation was in the top layer (0-15 cm) in the cotton field and in the deepest layer (30-60 cm) in the corn field (fig. 6). This might be a result of different factors, including differences in precipitation and irrigation, topographic and climatic factors, root distribution, water uptake, management practices, and soil hydraulic properties in the fields (Heathman et al., 2012; Liu et al., 2018; Shen et al., 2016).

The corn and cotton fields exhibited maximum spatial variation at the 30-60 cm and 0-15 cm depths, respectively, during the growing season. This might be due to topographic differences in the corn field and the cover crop in the cotton field. Cover crops help to hold water on the surface while minimizing soil evaporation (Heathman et al., 2012). Furthermore, topography had a significant effect on soil water at the 30-60 cm depth in the corn field (table 1) but no significant effect in the cotton field. Likewise, silt percentage in the cotton field had a significant effect on soil water at the 0-15 cm depth. Silt holds more water due to its finer texture. Hence, it can be concluded that the standard deviation was substantially dependent on soil water rather than soil depth during the growing season.

Furthermore, the impacts of different canopy structures, crop types, root water uptake, and atmospheric exposure of the surface, as well as precipitation and irrigation, help to

strengthen the spatial homogeneity and reduce the standard deviation (Liu and Shao, 2014; Vereecken et al., 2014; Wang et al., 2015; Zucco et al., 2014). One of the reasons for the spatial variability in both fields could be the no-tillage management in both fields, as well as the cover crop in the cotton field (Heathman et al., 2012). This is discussed in more detail later.

TEMPORAL STABILITY OF SOIL WATER

A representative of the mean soil water is required to understand the temporal stability of soil water in the two crop fields during the growing season. A relative difference approach was used to study the temporal stability. Therefore, statistics including $\bar{\delta}_i$, $\sigma(\delta_i)$, and ITS_i were evaluated and compared for all sensor locations in both fields. The $\bar{\delta}_i$ and $\sigma(\delta_i)$ values for each sensor location were ranked in ascending order with their respective depths (fig. 8). The $|\bar{\delta}_i|$ value ranged from 0.001 to 0.155 in the corn field and from 0 to 0.164 in the cotton field. The $\sigma(\delta_i)$ value ranged from 0.015 to 0.128 in the corn field and from 0.009 to 0.083 in the cotton field. The ITS_i value ranged from 0.018 to 0.180 in the corn field and from 0.010 to 0.165 in the cotton field.

Tables 2 and 3 list the Spearman's rank correlation (r_s) values between different soil depths to evaluate the spatial correlations for $\bar{\delta}_i$, $\sigma(\delta_i)$, and ITS_i . Table 2 shows that r_s was significant between all soil depths at a significance level of 0.05 in the corn field. In the cotton field, r_s was not significant between the 0-15 cm and 30-60 cm depths for $\bar{\delta}_i$ and between all soil depths for ITS_i . All the significant correlations were positive for all soil depths. Although the spatial patterns were not consistent with soil depth in both fields, the 0-15 cm and 15-30 cm depths showed stronger correlations compared to other depths in the corn field.

Values of $\bar{\delta}_i$ that are close to zero and lower values of $\sigma(\delta_i)$ and ITS_i indicate temporal stability of soil water. These ranges sometimes mislead the results to define a location as temporally stable that could represent the spatial mean soil water (Dari et al., 2019). Therefore, to verify if a representative location was temporally stable, additional statistical criteria were adopted, including r_s , R^2 , RMSE, IA, KGE, and NSE, between a representative location and the spatial mean soil water (table 4). In the corn field, sensor 8 (S8) was found to be temporally stable at all soil depths (figs. 8a to 8c) and can therefore be used as a representative sensor location to estimate mean soil water in the corn field during the growing season (figs. 9a to 9c). This may be due to soil hydraulic properties at S8 that are similar to the mean soil hydraulic properties, combined with soil homogeneity and soil water in the field (Yetbarek and Ojha, 2020). The adopted statistics verify that S8 is the best representative sensor and also support the relative difference approach (table 4). Sensors S7, S9, S18, S22, S23, and S30 were always dry, while sensors S4, S5, S10, S12, S13, S21, and S24 were always wet for the entire soil profile. This variation in dryness and wetness indicated soil heterogeneity from location to location and homogeneity within the entire soil profile at particular locations. Topographic differences can also cause such

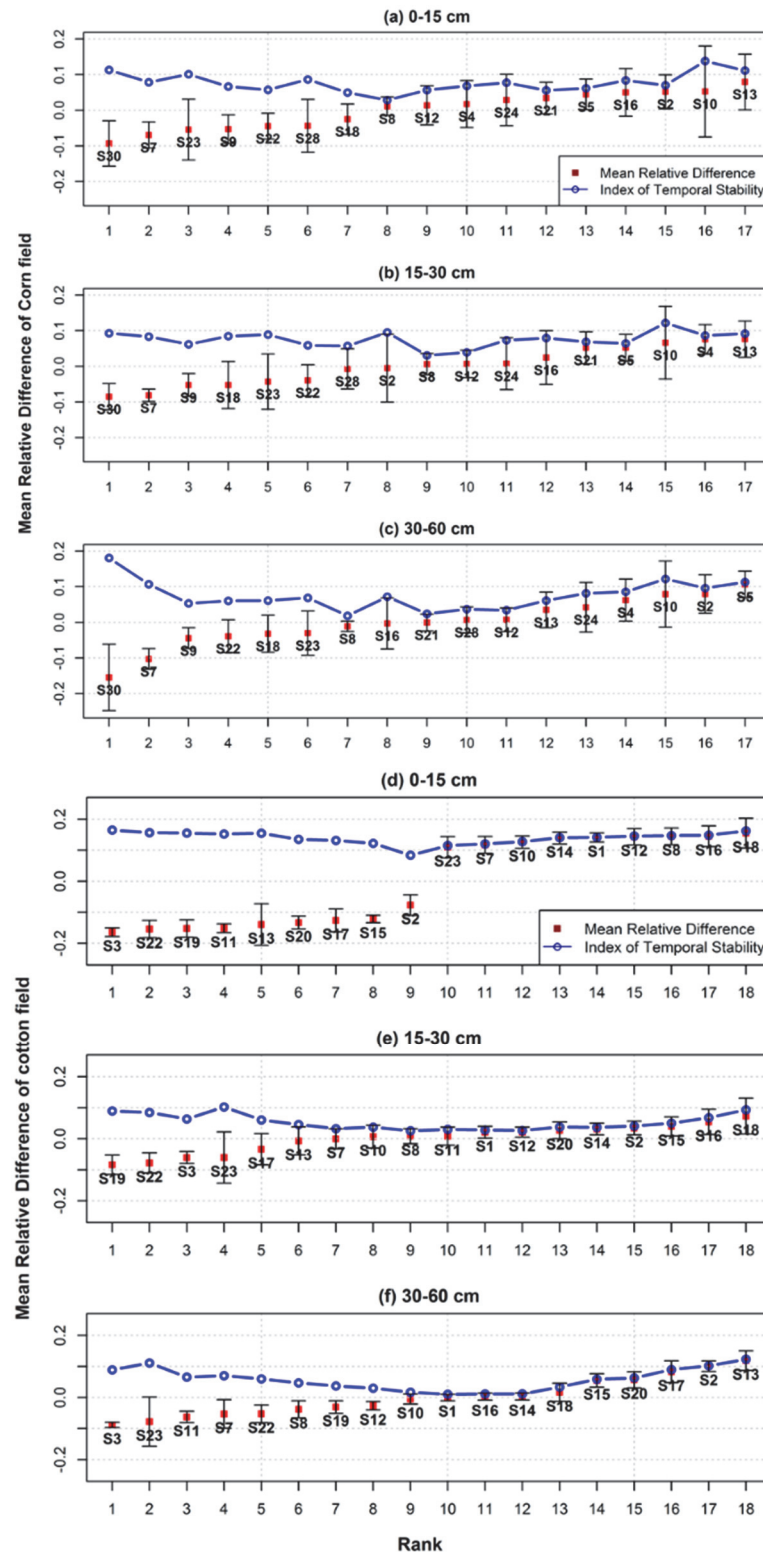


Figure 8. Plots of rank ordered mean relative difference and index of temporal stability of soil water in the (a-c) corn field and (d-f) cotton field. Error bars show standard deviations in relative difference.

variations in the dryness and wetness of sensors. However, the rest of the sensors were in a transitioning phase of dry and wet at different depths. Sensor S30 was the driest at all soil depths, while sensor S13 was the wettest in the upper two soil layers during the growing season.

In the cotton field, sensor S1 was temporally stable at all depths during the growing season (figs. 8d to 8f), and it can be used as a representative sensor location to estimate the mean soil water in the cotton field (figs. 9d to 9f). Although S1 showed $\bar{\delta}_i$ and ITS_i values of 0.14 at the 0-15 cm depth,

Table 2. Spearman's rank correlation coefficients between soil depths in corn field (all correlations are significant at 0.05).

Statistics	Soil Depth (cm)		
		15-30	30-60
$\bar{\delta}_i$	0-15	0.87	0.82
	15-30	-	0.81
$\sigma(\delta_i)$	0-15	0.68	0.65
	15-30	-	0.61
ITS _i	0-15	0.65	0.58
	15-30	-	0.72

Table 3. Spearman's rank correlation coefficients between soil depths in cotton field. Asterisks (*) indicate non-significance at 0.05.

Statistic	Soil Depth (cm)		
		15-30	30-60
$\bar{\delta}_i$	0-15	0.66	0.17*
	15-30	-	0.50
$\sigma(\delta_i)$	0-15	0.70	0.60
	15-30	-	0.69
ITS _i	0-15	0.23*	-0.19*
	15-30	-	0.26*

it provided the most accurate spatial mean soil water at all depths. Sensor S1 showed temporal stability at deeper soil layers in the cotton field (table 4). However, according to the relative difference approach, no sensor showed a perfect zero value for $\bar{\delta}_i$ in the top soil depth (fig. 8d), which suggests that no sensor was temporally stable at the 0-15 cm depth. The IA and NSE values confirmed this lack of

temporal stability at the 0-15 cm depth (table 4). However, r_s , R^2 , RMSE, and KGE presented convincing evidence to consider S1 a representative sensor at the 0-15 cm depth. Therefore, S1 can be considered a temporally stable location at all soil depths, and it can be used as a representative sensor location to estimate mean soil water in the cotton field during the growing season. Sensors S3, S19, and S22 were always dry, while sensor S18 was always wet for the entire soil profile during the growing season. Sensor S3 was the driest at the 0-15 and 30-60 cm depths, while sensor S18 was the wettest in the upper two soil depths in the cotton field during the growing season (figs. 8d to 8f). Other sensors similar to S1 were also able to produce the mean soil water (figs. 9d to 9f), but the statistical parameters were not as good as for S1.

Any sensor that has statistics similar to S8 in the corn field and similar to S1 in the cotton field, or a sensor that can estimate mean soil water, can be used to represent mean soil water using an offset correction to serve as a temporally stable sensor (Heathman et al., 2012; Martínez-Fernández and Ceballos, 2005; She et al., 2012, 2015). Therefore, according to the $\bar{\delta}_i$, $\sigma(\delta_i)$, ITS_i, r_s , R^2 , RMSE, IA, KGE, and NSE values, because the goal was to find the best representative sensor location to estimate mean soil water in crop fields, S8 in the corn field and S1 in the cotton field were selected as temporally stable sensors during the growing season. This decision might help the farmers in making decisions to reduce

Table 4. Summary of temporal stability statistics between field mean soil water and representative sensor.

Statistic	Corn Field (S8)			Cotton Field (S1)		
	0-15 cm	15-30 cm	30-60 cm	0-15 cm	15-30 cm	30-60 cm
r_s	0.84	0.75	0.95	0.95	0.98	0.96
R^2	0.76	0.85	0.97	0.90	0.97	0.94
RMSE	0.007	0.009	0.007	0.053	0.011	0.004
IA	0.9	0.93	0.99	0.42	0.96	0.98
KGE	0.76	0.68	0.94	0.84	0.80	0.89
NSE	0.70	0.79	0.95	-9.07	0.85	0.93

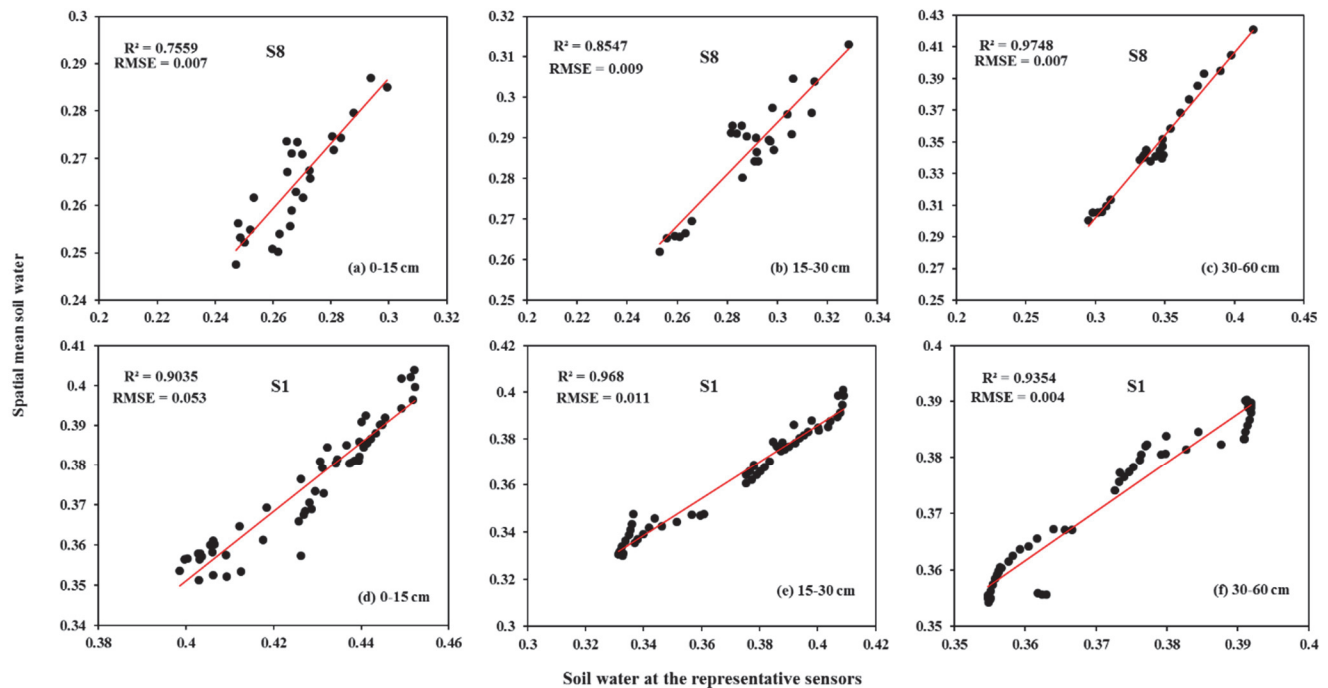


Figure 9. Relation between spatial mean soil water and representative sensors in the corn field at (a) 0-15 cm, (b) 15-30 cm, (c) 30-60 cm depths, and in the cotton field at (d) 0-15 cm, (e) 15-30 cm, and (f) 30-60 cm depths during the growing season.

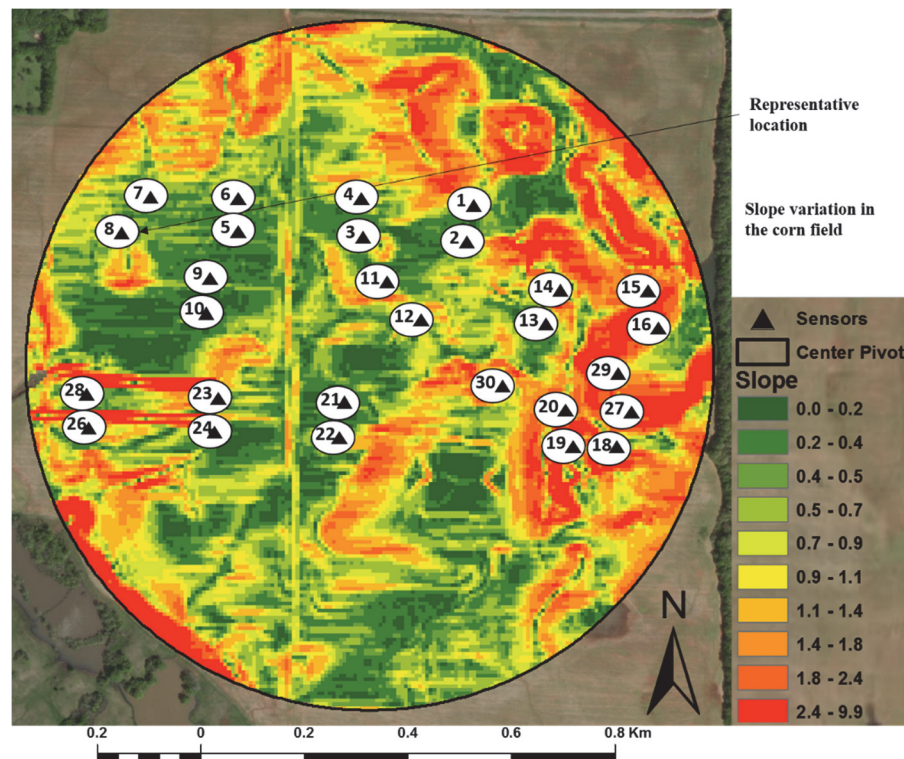


Figure 10. Location of representative sensor (S8) in corn field with respect to other sensors in the field.

Table 5. Properties of representative sensors (S1) in cotton field with respect to field average.

Soil Depth (cm)	Sensor S1		Field	
	Silt (%)	EC	Silt (%)	EC
0-15	53.8	NA	54.4	NA
15-30	52.65	18	47.55	16
30-60	49.48	44.54	54	51

costs and labor during the growing season. The temporal stability increased with soil depth in the crop fields (table 4). Sensor S8 in the corn field was located at an elevation of 173 m. The properties of the representative locations (S8 in corn and S1 in cotton) are shown in figure 10 and table 5.

FACTORS RESPONSIBLE FOR SOIL WATER VARIABILITY

The mean soil water was affected differently in the two crop fields. Spearman's rank correlation was estimated between soil water and the parameters considered responsible for the soil water variability. The topographic variation in a crop field can be a factor responsible for soil water variability, especially in deep soil layers. A high slope causes flow in a lateral direction, as surface or subsurface runoff, and reduces infiltration (Huat et al., 2006). The topographic attributes in the corn field showed an impact on soil water variation during the growing season. Sensor S30 was located on a high slope that could have led to surface or subsurface runoff and reduced infiltration, resulting in the driest 30-60 cm soil layer. However, sensor S5 was located on a low slope, which could have resulted in the wettest 30-60 soil layer in the corn field. Most of the weather parameters showed a significant positive correlation with soil water in the corn field but a significant negative correlation with soil water in the cotton field. This difference might be due to differences in the plant canopies during the growing season, in addition to

variation in topography, climate, root system, soil properties, and size of the fields.

ET_c was significantly poorly correlated (-0.33) with soil water at the 15-30 cm depth in the corn field (table 6). However, ET_c showed a significant moderate correlation (-0.60 to -0.67) with soil water at all soil depths in the cotton field. ET_c showed a significantly higher correlation with soil water at the 15-30 cm depth in both fields, which suggests a great influence of ET_c on soil water at the 15-30 cm depth as compared to the other soil depths. This could be because cotton has a deeper root system (>90 cm) for water uptake as compared to corn (50 cm) (Liu et al., 2018; Wang et al., 2010). Studies have shown that corn and cotton have shallow and deep root systems, respectively, for water uptake (Liu et al., 2018). Root water uptake can be described using the root depths for both crops during the growing season, which affects the soil water variation. It has been determined that root water uptake increases with root depths greater than 90 cm during the growth stages in cotton. However, in corn, the depth of root water uptake increases to 50 cm from the jointing stage to flowering but decreases in the full maturity stage. Therefore, the soil water variation between the two crops could be due to different root water uptake in the fields during the growing season. In addition to the root system, cotton has a small and light canopy during the growing stages as compared to corn, which has a long and dense canopy during the growing stages.

It has also been observed that no-till and cover crops help restore soil water in the soil (Heathman et al., 2012). Both fields were under no-till management before seeding. However, the cotton field was planted with a cover crop before the growing season, and no cover crop was planted in the corn field. Therefore, the variation in soil water in the fields

Table 6. Spearman's rank correlation between spatial mean soil water and meteorological parameters.^[a]

Factor	Corn Field			Cotton Field		
	0-15 cm	15-30 cm	30-60 cm	0-15 cm	15-30 cm	30-60 cm
ET _c (cm)	-0.08	-0.33*	-0.31	-0.63*	-0.67*	-0.60*
Mean air temperature (°C)	0.35*	0.41*	0.42*	-0.46*	-0.55*	-0.53*
Solar radiation (W m ⁻²)	0.03	0.24	0.29	-0.37*	-0.42*	-0.39*
accGDD (°C)	-0.19	-0.67*	-0.95*	0.39*	0.31*	0.27*

^[a] Asterisks (*) indicate significance at 0.1.

can also be due to cover crop management practices (Heathman et al., 2012). The mean air temperature and solar radiation showed stronger correlations in deeper soil layers in both fields. The accumulated growing degree days (accGDD) showed an increasing negative correlation with increasing depth in the corn field and a decreasing positive correlation in the cotton field. This was due to the different root distributions of the crops, as mentioned previously.

Different topographic attributes, such as elevation and slope, were considered to evaluate the effects of these parameters on soil water variations in both fields during the growing season. The sand, silt, and clay percentages and the soil EC were also considered for their effect on soil water. In the corn field, elevation was significantly correlated with soil water at the 30-60 cm depth (table 1), which shows the impact of topography on soil water variation in the deepest soil layer. No other field attributes (slope and sand, silt, and clay percentages) were significantly correlated with soil water at any soil depth in the corn field. However, the silt percentage at 0-15 cm and EC at 15-30 cm showed significant correlations with soil water in the cotton field. This might be due to the cover crop in the cotton field, which held water at the surface, along with the silt. No other field attributes (elevation, slope, sand, and clay) were significant in the cotton field. The variation in r_s was found to be due to difference in elevation in the corn field and to soil properties in the cotton field (table 1).

SUMMARY AND CONCLUSIONS

This study of spatiotemporal variability in soil water at three depths (0-15 cm, 15-30 cm, and 30-60 cm) was performed in two crop fields (corn and cotton) during the 2018 growing season in the TVR in northern Alabama. It can be concluded that soil water can be a key variable for determining uniform irrigation in different croplands. Soil water can

be used to optimize the amount, time, and rate of irrigation in crop fields during the growing season. The standard deviation did not increase with increasing soil water (as mentioned by Shen et al., 2016, for all soil depths in croplands). The soil water variability in the two fields was substantially dependent on soil water in the different soil layers. Therefore, soil water in the intermediate soil layer can play a vital role in soil water variability in crop fields due to irrigation from the upper soil layer and suction from the bottom soil layer.

Farmers can reduce the number of sensors installed in their fields to save time, cost, and labor during the growing season, and farmers can schedule precise uniform irrigation (right amount and right time) considering the entire soil profile. The temporal stability increased with soil depth in the crop fields irrespective of the irrigation application and the factors affecting soil water variation during the growing season. However, dry, wet, and temporally stable locations need to be monitored to understand the spatial variation in soil water in the entire field. After knowing the soil water variability, farmers need to decide the number of sensors based on their budget. The number of sensors can also be determined based on statistical criteria. Statistical techniques can reduce the number of sensors, but the cost has to be paid by the farmer. Therefore, we suggest that farmers decide the number of sensors based on their budget.

Using the standard deviation through implicit relationships at absolute errors of 5% and 10%, the number of sensors was estimated for both crop fields. Figure 11 shows that a maximum of ten sensors in the corn field and six sensors in the cotton field would be required to determine the mean soil water with 5% absolute error. Topography can be an influential factor in a field similar to the corn field (topographic variations), which needs to be considered, but not in a field similar to the cotton field (almost flat). The number of sensors can also be related to crop types with similar root distributions and canopy structures.

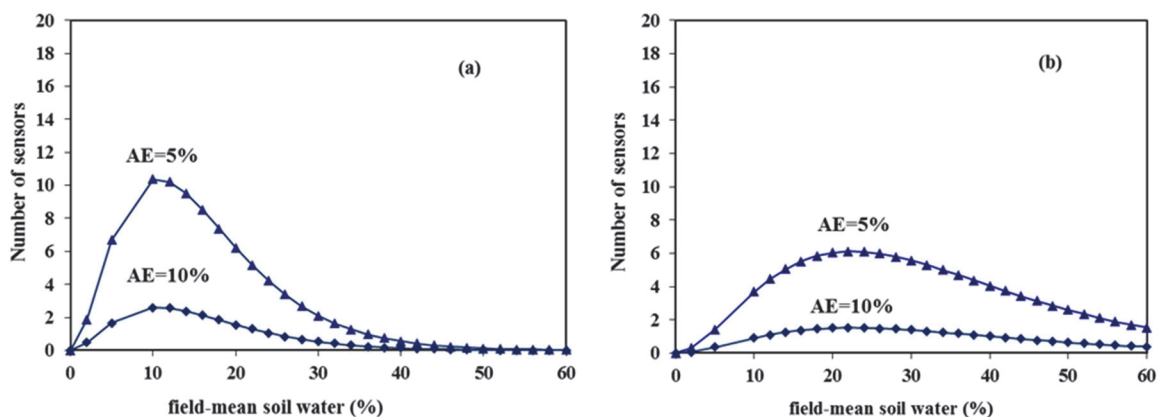


Figure 11. Number of sensors required in the (a) corn and (b) cotton fields.

This study was conducted at various depths in two crop fields during a wet season, based on the historical average precipitation. Irrigation is expensive. Therefore, we need to investigate the spatiotemporal variability of soil water in different growing seasons and in different crops over multiple years to help farmers adopt irrigation while reducing the associated cost. We focused on 60 cm of soil depth in this study. However, to determine the full soil water balance based on the entire root depth of a crop, sensors should be installed in soil layers deeper than 60 cm to determine if the measurement of soil water in deeper soil layers (below 60 cm) can improve water use efficiency.

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